

PROFESSIONAL MACHINE INSPECTORS LIMITED (PMI)

Professional machine inspectors limited (PMI) is an engineering firm based in Nairobi – Kenya. PMI was incorporated in November 2009 and specializes in the field of condition based maintenance programs (CBM) serving different industries including Hydro power generating companies. We specialize in the following CBM programs:

1. Vibration analysis
2. Thermography
3. Precision laser alignment
4. Onsite dynamic balancing
5. Trainings
6. Machine overhauls

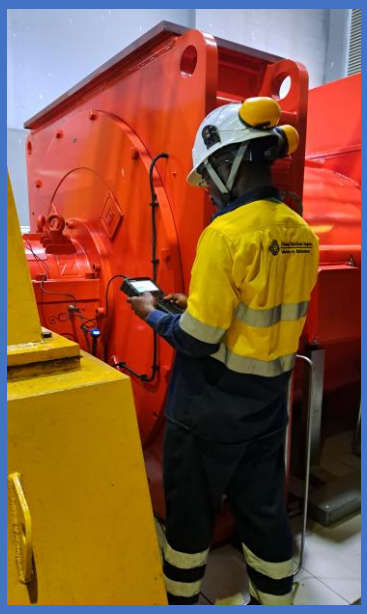
PMI strives to offer the listed CBM solutions closer to small, medium and large sized hydro power plants with critical machines in their generating process, thereby helping them eliminate expensive breakdowns. Our company is run by a competent team of practitioners in Engineering, Project Management, Marketing and Finance that is guided by the mission.

“To be a world class provider in Condition Based Maintenance services and products”.

Professional Machine Inspectors Ltd.
Adding value to Maintenance

SERVICES

1. Vibration Analysis



Vibration analysis is a powerful tool for detecting both electrical and mechanical issues in machinery. It helps identify potential problems early, preventing catastrophic disruptions to operations. Insights from vibration analysis guide planning teams to schedule timely corrective actions. This service is ideal for critical rotating equipment, such as Generators and turbines.

2. Thermography



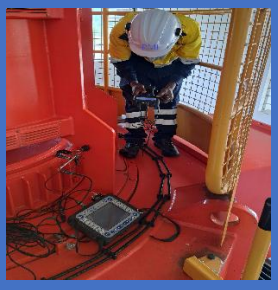
We use advanced thermal images and experienced personnel to conduct infrared analysis on critical plant equipment. This technique detects hotspots and faults, helping prevent costly breakdowns or operational disruptions. Thermography is ideal for electrical panels, transformers, contactors, power distribution boards, substations, e.t.c.

3. Precision Alignment



We utilize advanced laser alignment tools to ensure precise alignment, reducing maintenance costs and power consumption by up to 17%. Our services cover both horizontal and vertical hydro power plants machine train alignment. Laser alignment is especially critical after decoupling machine trains or during shutdowns.

4. Onsite Dynamic Balancing



PMI offers on-site dynamic balancing, eliminating the need to transport machines to the workshop. We perform both single and duo-plane balancing for the runner. The runner can become unbalanced due to buckets wear and tear of the buckets as well as cavitation, which may lead to excessive vibration

and strain on components. Our balancing solution restores optimal performance efficiently.

5. Training



PMI provides specialized training in Condition Based Maintenance (CBM) and vibration analysis, equipping maintenance teams with the skills to monitor, diagnose and prevent equipment failures. Our modules focus on data collection, analysis, and predictive maintenance, helping clients minimize downtime, extend equipment lifespan, and boost operational efficiency with cutting-edge industry practices.

6. Machine Overhauls







PMI restores equipment to optimal condition through overhauls performed by skilled engineers and technicians using the right tools and adhering to engineering standards. Post-overhaul, we monitor performance using technologies like vibration analysis and thermography to ensure peak efficiency.